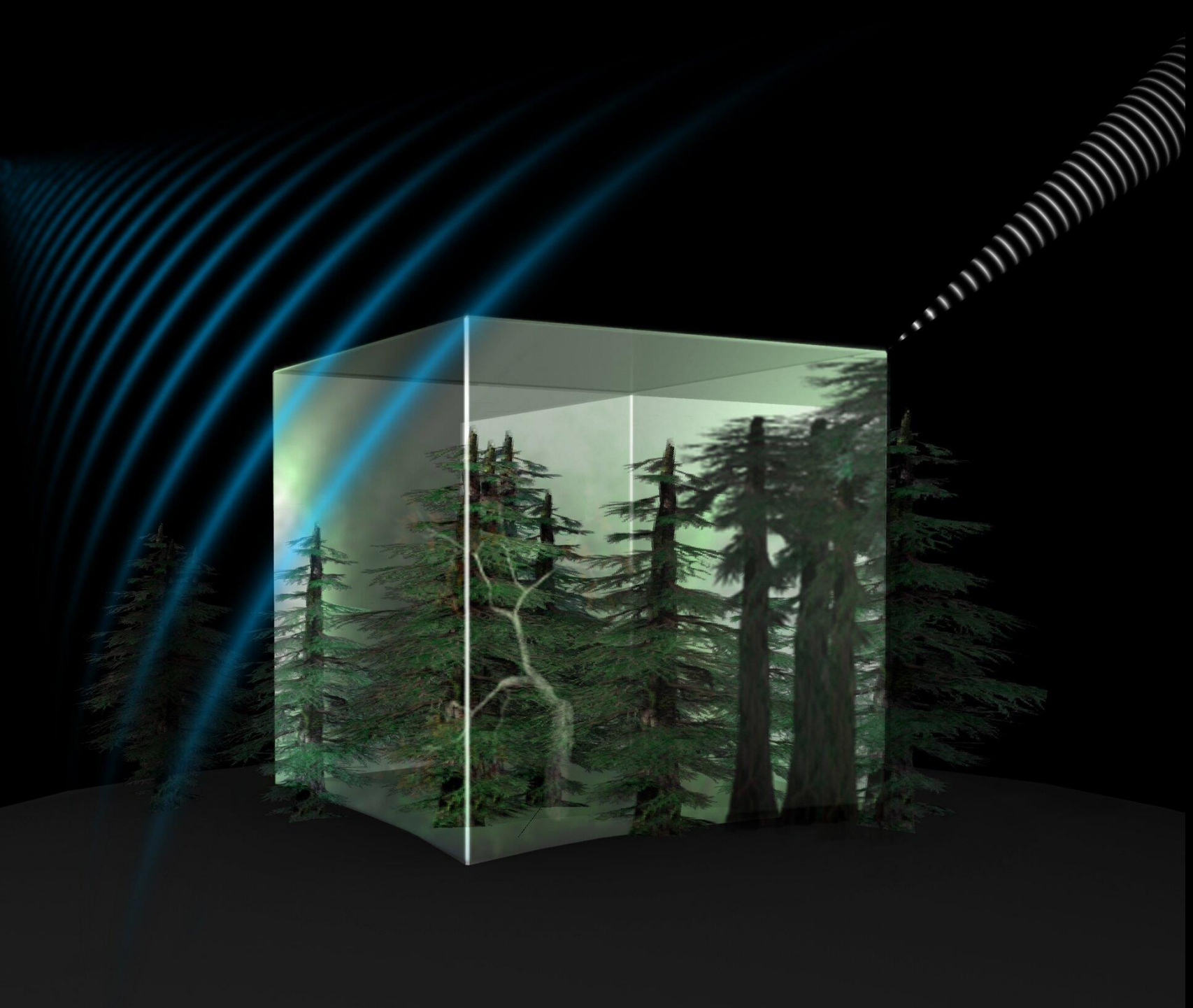
Land Ice Structures with P-band SAR

Georg Fischer[1], Giuseppe Parrella[1], Matteo Pardini[1], Kostas Papathanassiou[1], Irena Hajnsek[1/2]

georg.fischer@dlr.de

[1] Microwaves and Radar Institute, DLR Oberpfaffenhofen
[2] Institute of Environmental Engineering, ETH Zurich

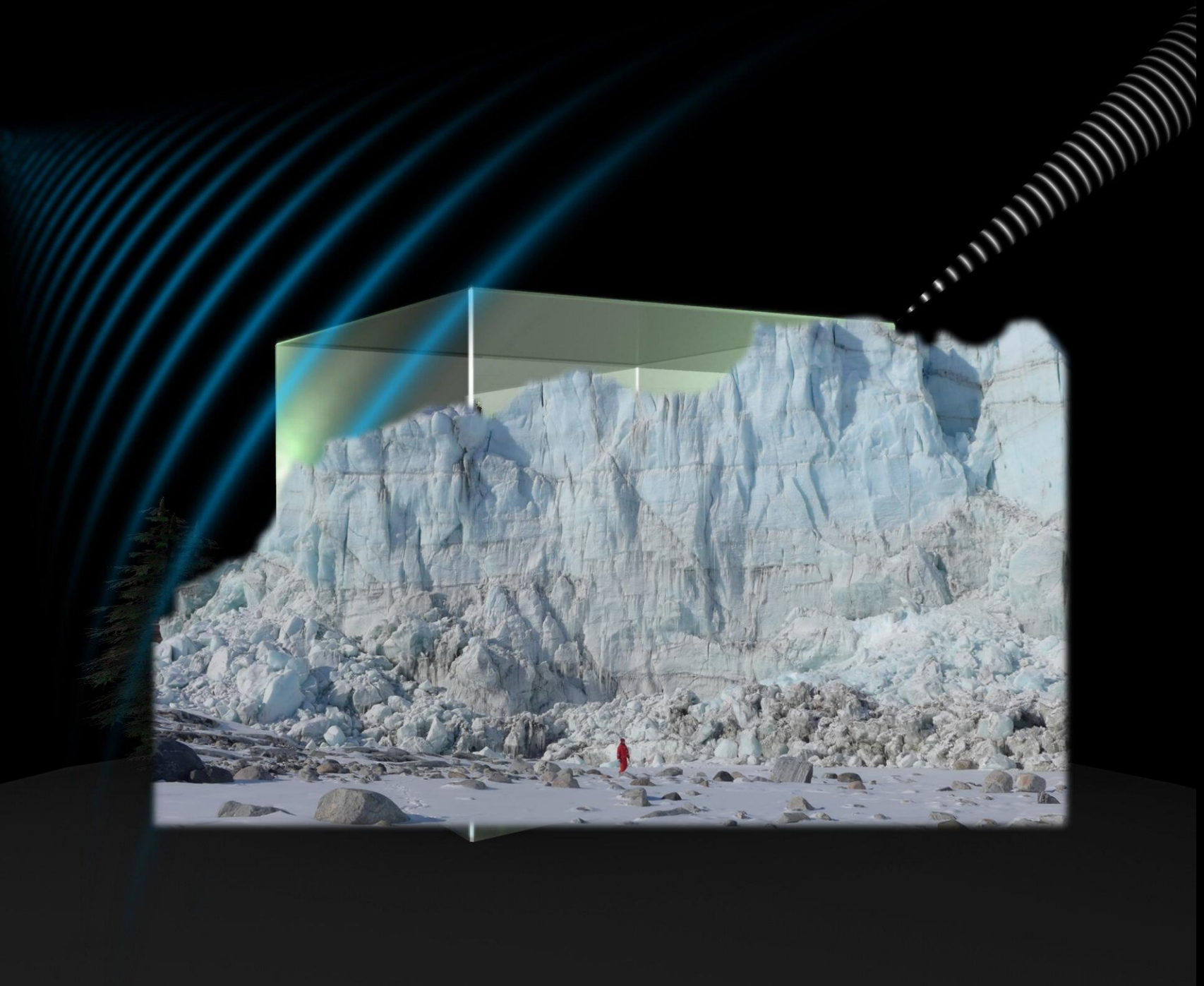


Why Land Ice Structure?

Why Synthetic Aperture Radar?

Why PolSAR and TomoSAR?

Why P-band?



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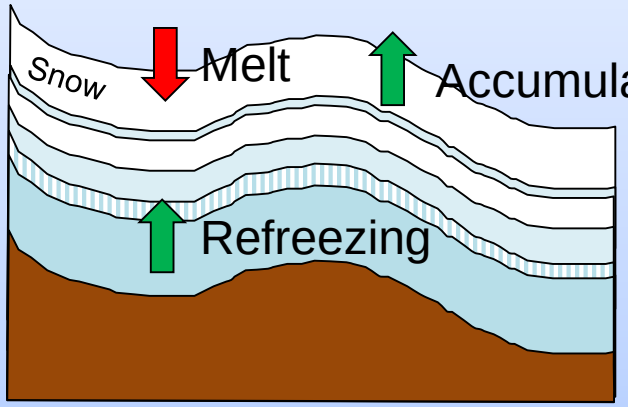
Why Land Ice Structure?

Snow ≠ Snow ≠ Firn ≠ Ice

Geophysical Subsurface Processes



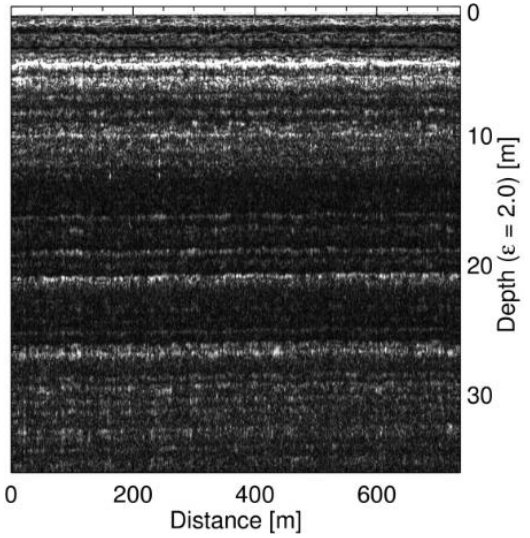
Volume Change

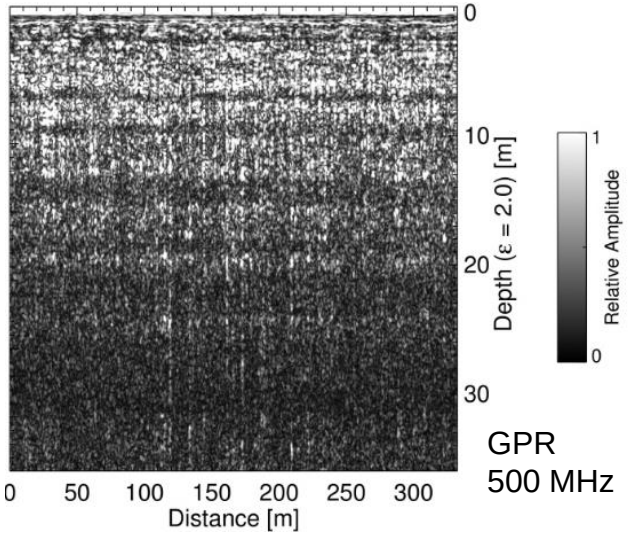
≠

Mass Change

Density changes due to refreezing
→ Uncertainty in mass balance estimations
[IPCC AR5, 2013; Parry et al., 2007]

In situ ground penetrating radar: vertical scattering profiles





GPR
500 MHz

Less melting
Distinct layers
Lower density

⟺

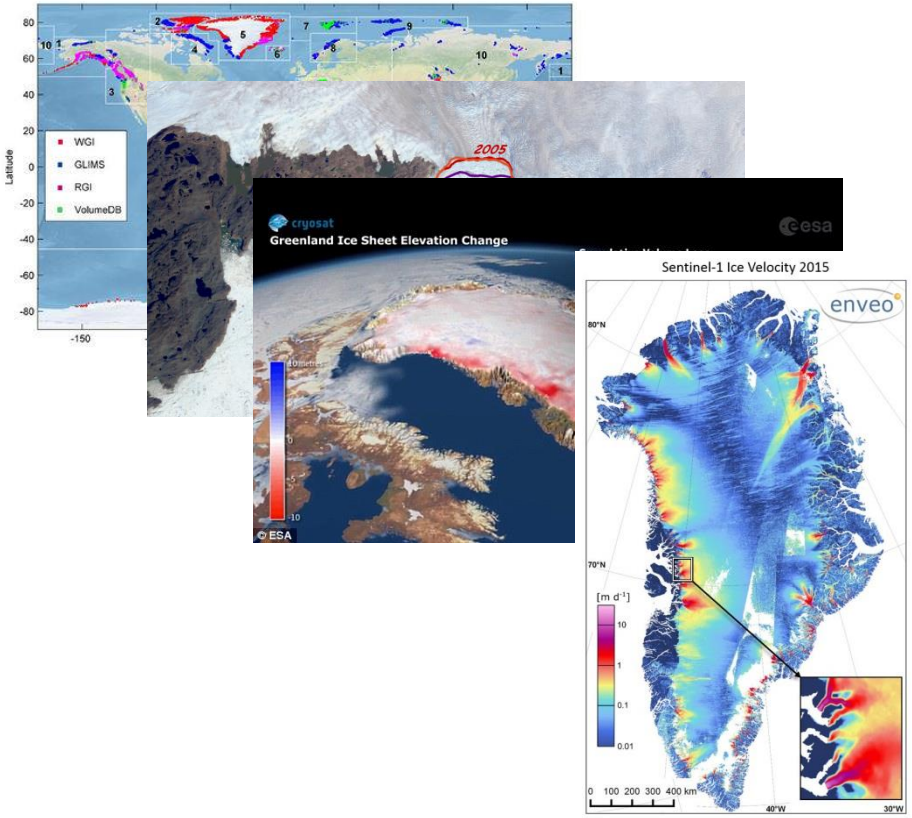
More melting
and ice inclusions
Higher density

Why Synthetic Aperture Radar?

Remote Sensing of Glaciers – State of the Art

- Location
- Size
- Thickness
- Velocity

→All established/operational

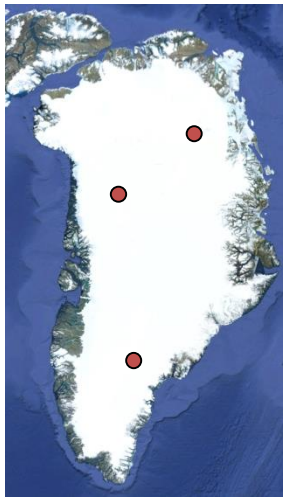


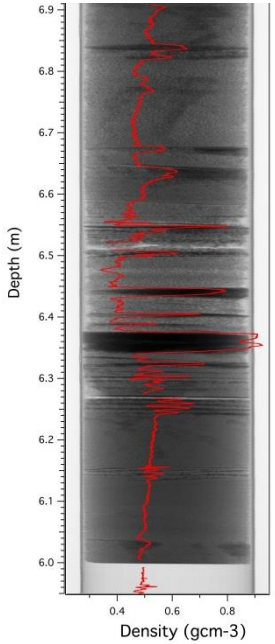
Open Question:

What's changing below the surface?

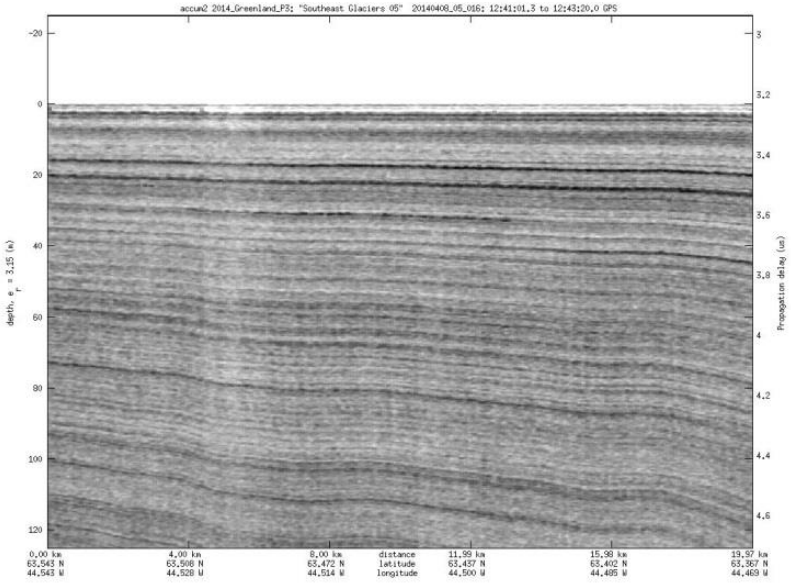
- Density?
- Refreezing of melt water?
- Layer structure?
- …

Field Measurements

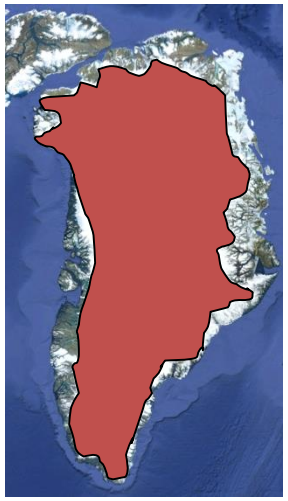


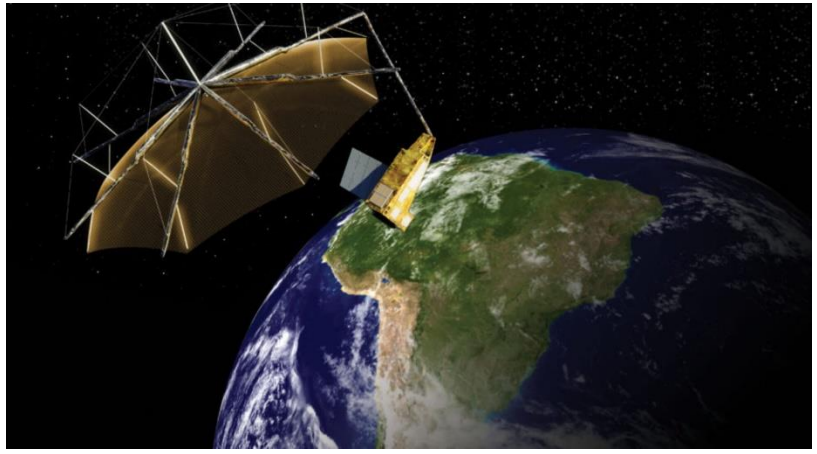


Airborne Radar Sounder



Spaceborne SAR





Why Synthetic Aperture Radar?

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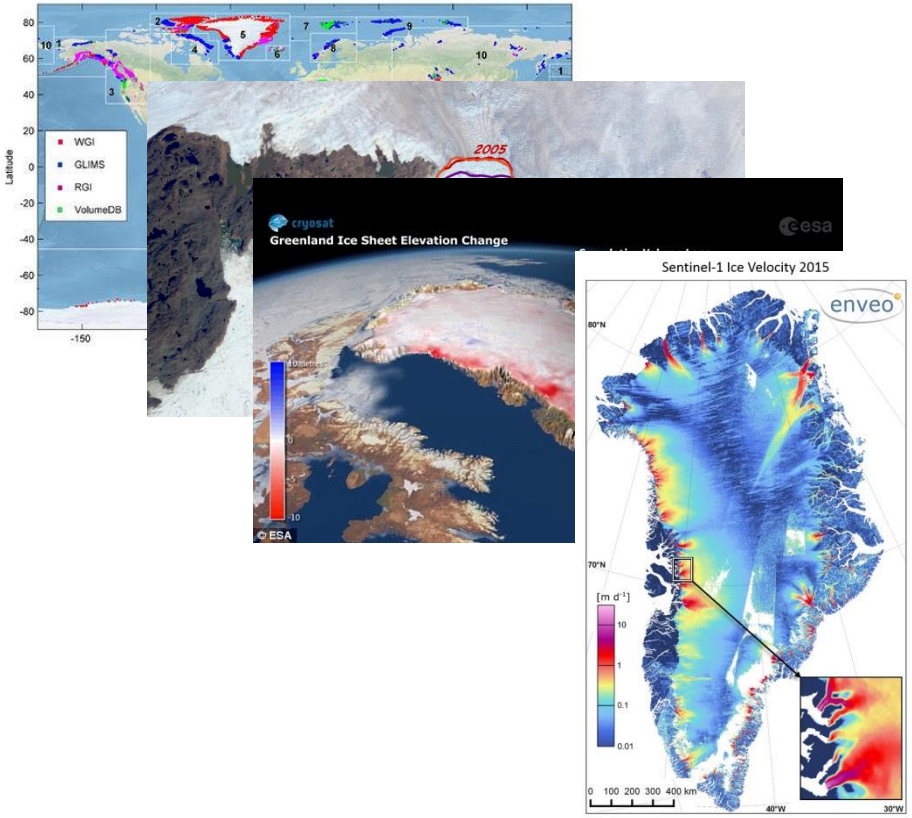
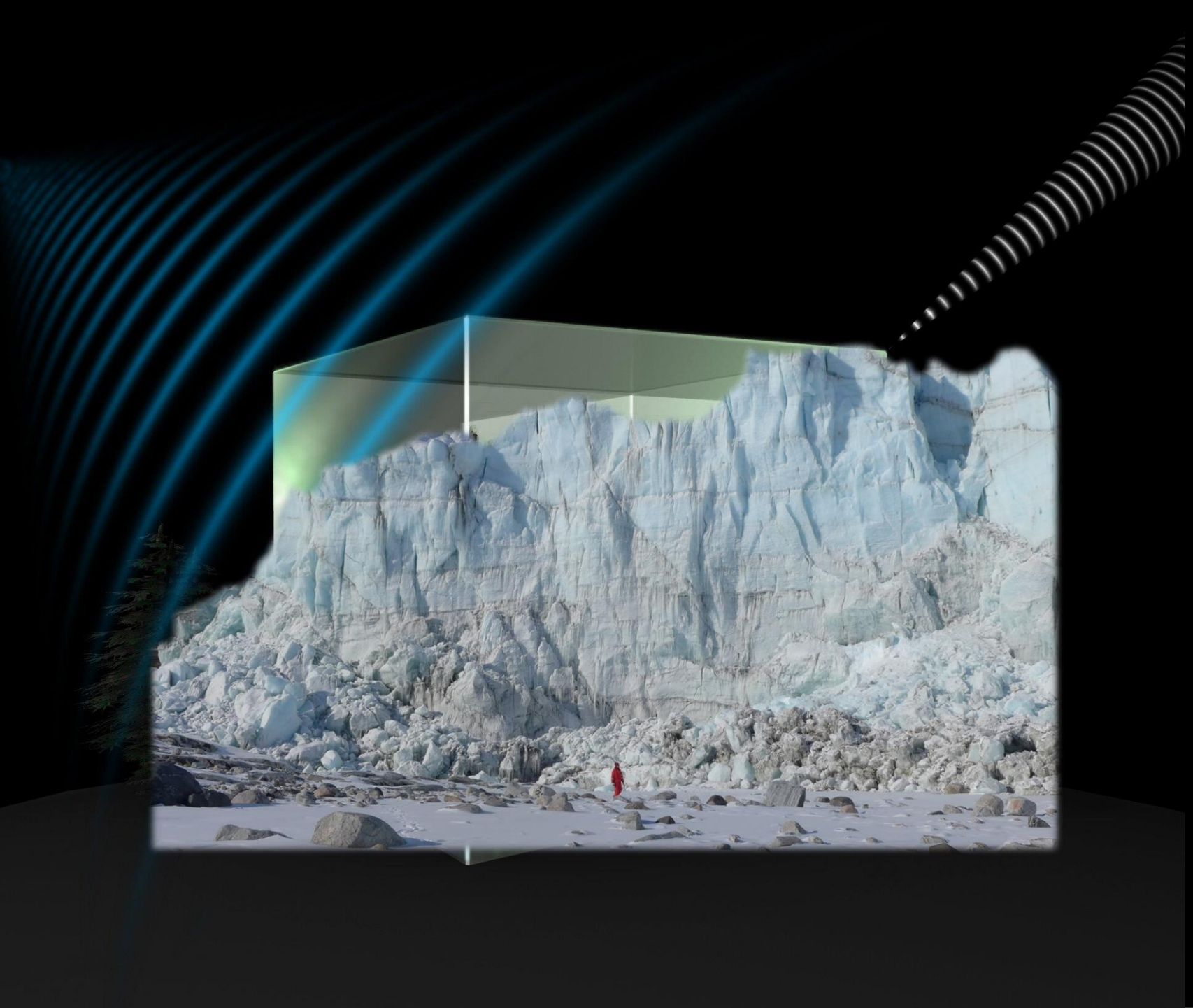


Open Question:

What's changing below the surface of Greenland?

- Density?
- Refreezing of melt water
- Layer structure?
- …

⇨ **long wavelength Pol-InSAR !**



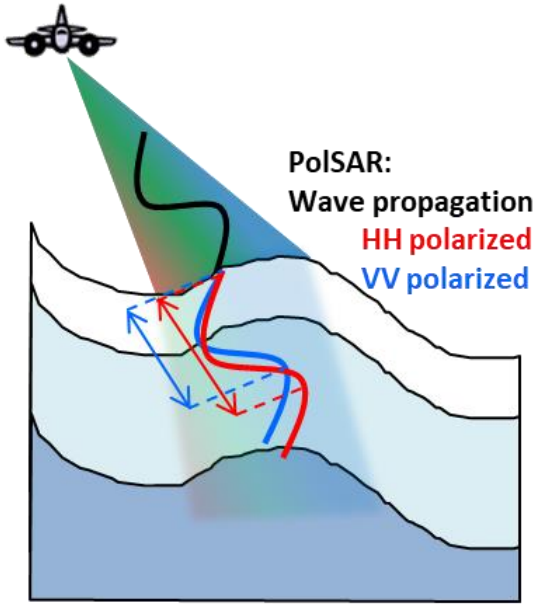
Why Land Ice Structure?

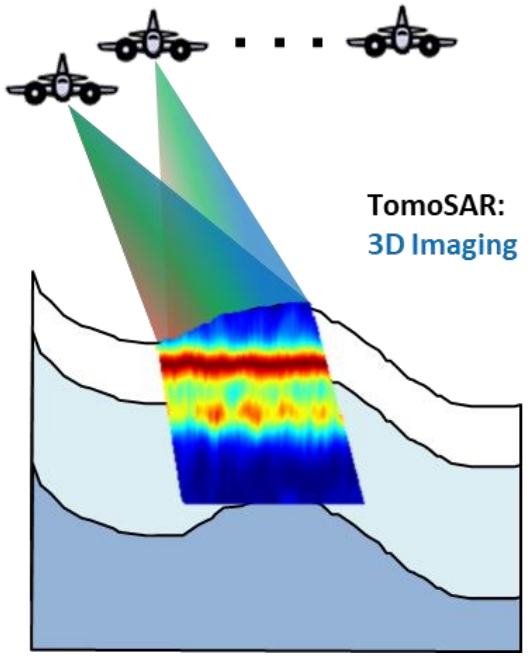
Why Synthetic Aperture Radar?

Why PolSAR and TomoSAR?

Why P-band?

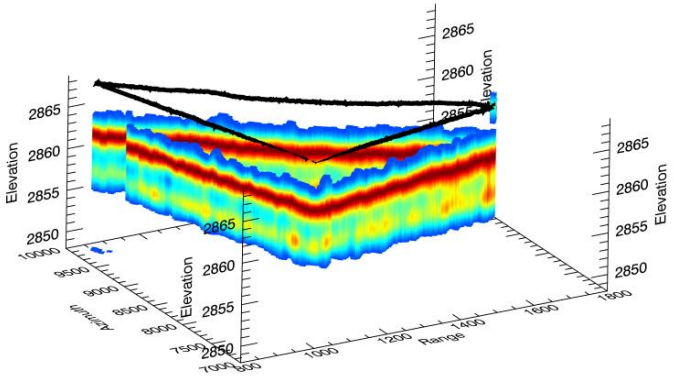
Why PolSAR and TomoSAR?





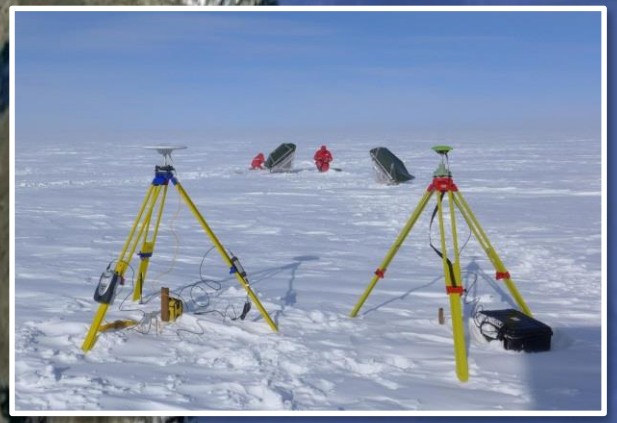
PolSAR → Scattering mechanisms
Propagation properties

TomoSAR → Subsurface scattering structure

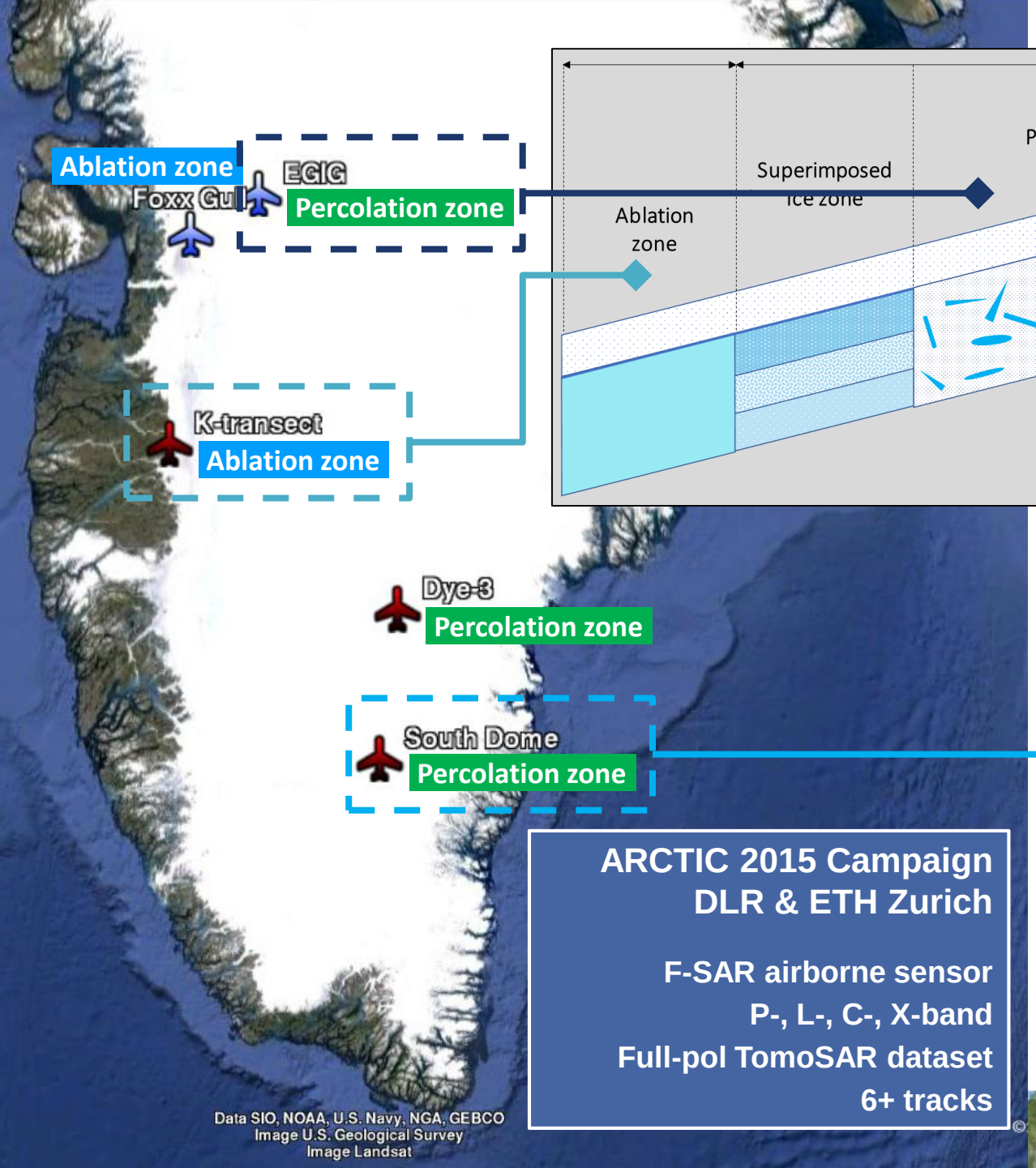


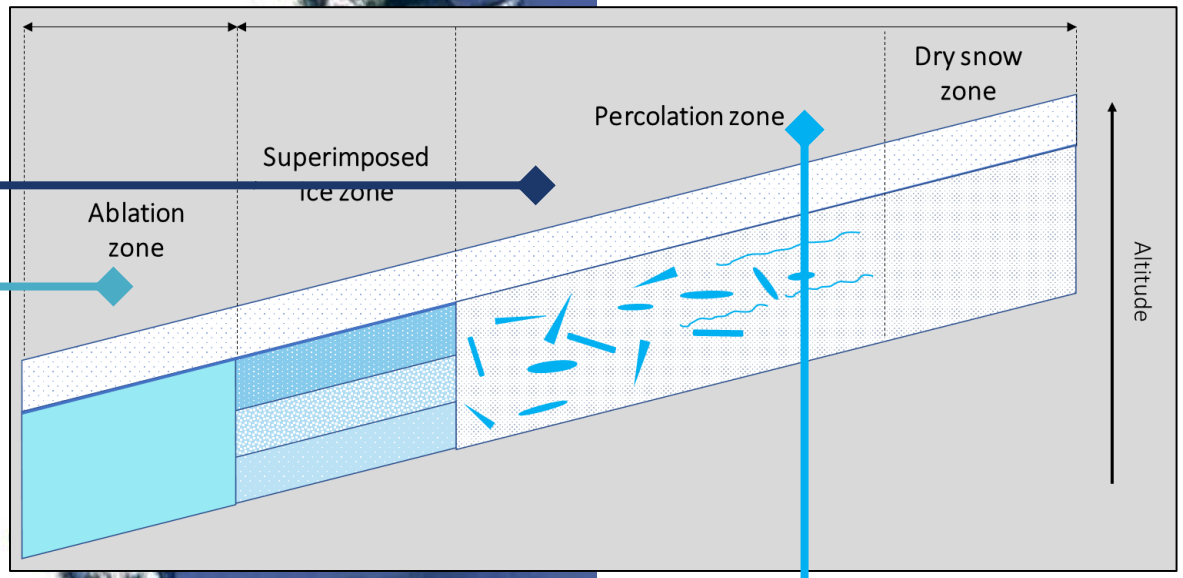
Why P-band?

Signal penetration depth is key!



Pictures by Silvan Leinss



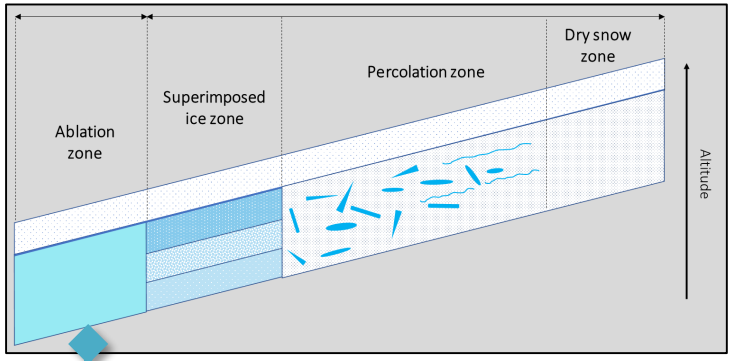


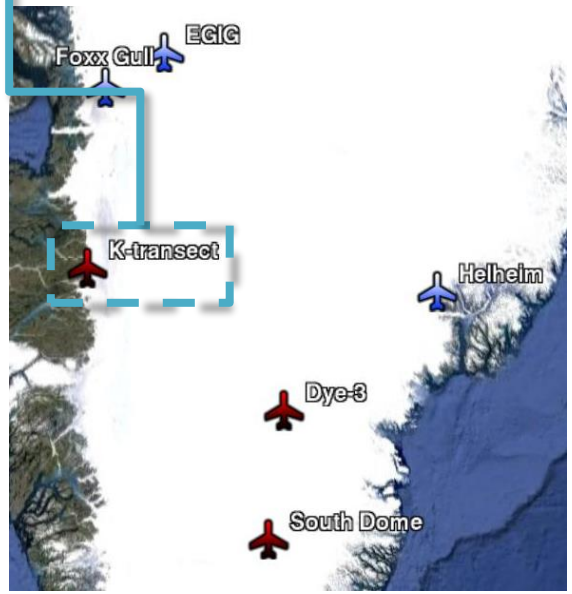
ARCTIC 2015 Campaign
DLR & ETH Zurich

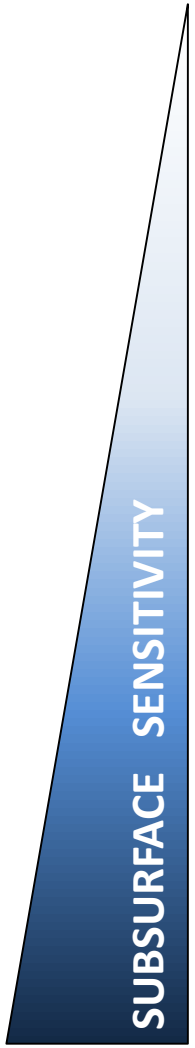
F-SAR airborne sensor
P-, L-, C-, X-band
Full-pol TomoSAR dataset
6+ tracks

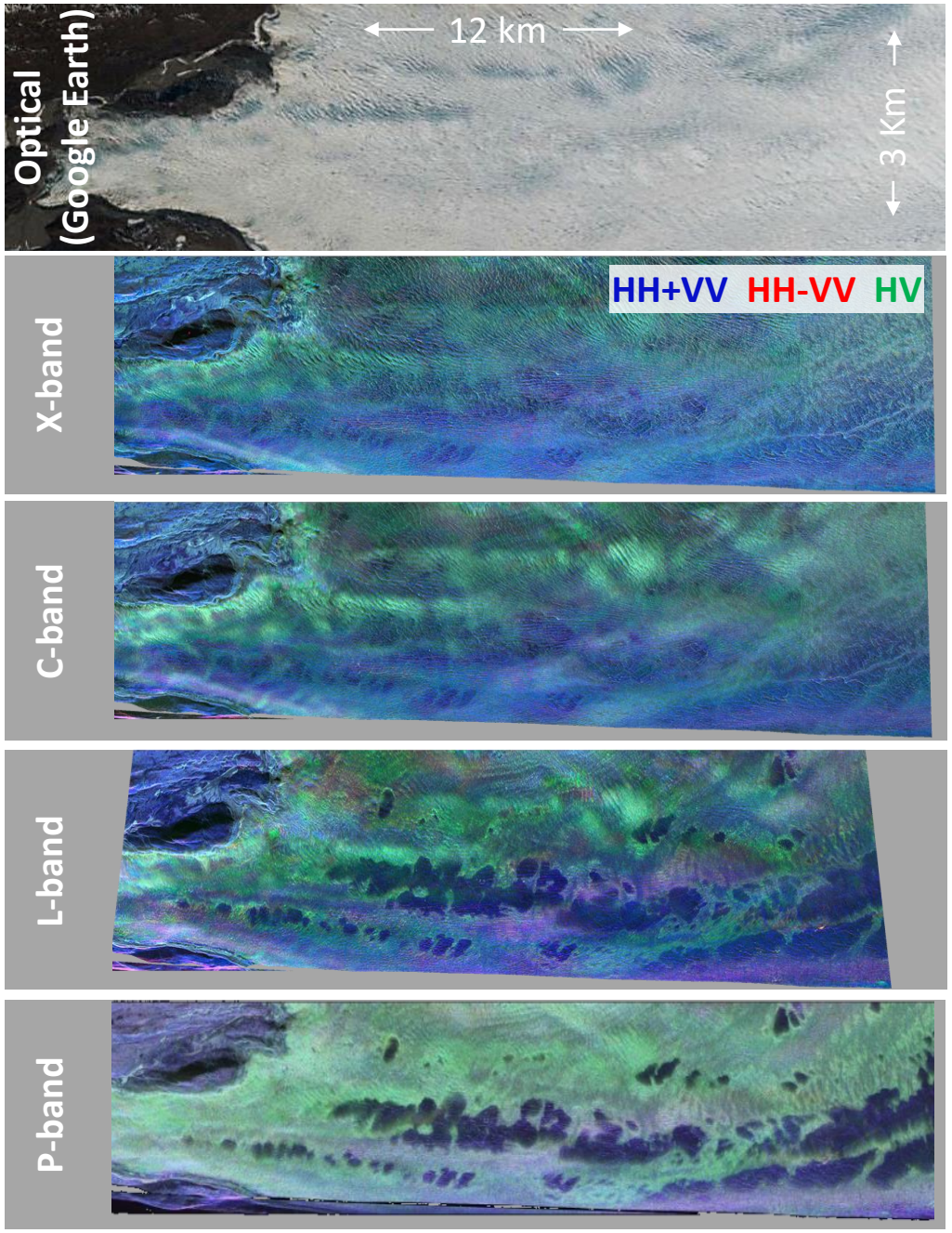
DLR

Why P-band PolSAR?

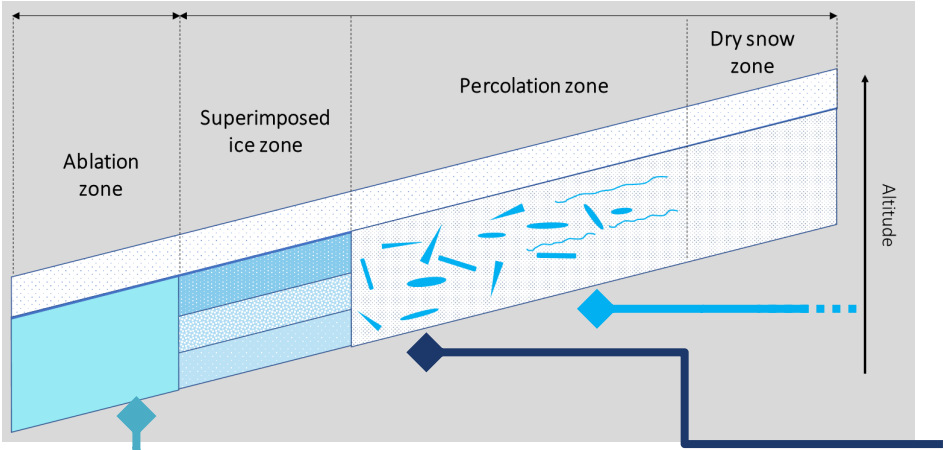






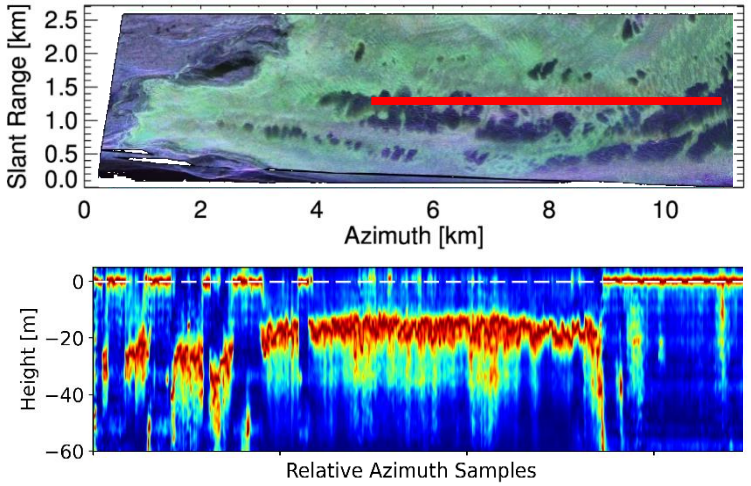


Why P-band TomoSAR?



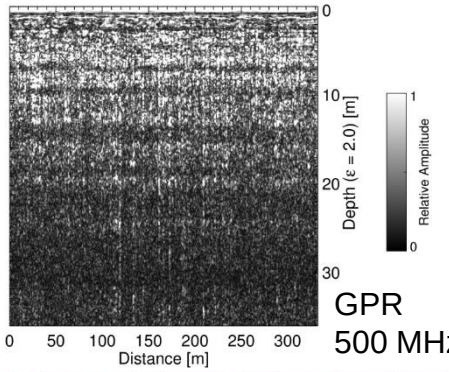
K-Transect

solid glacier ice



EGIG T05

More melting
Uniform scattering
Higher density



GPR
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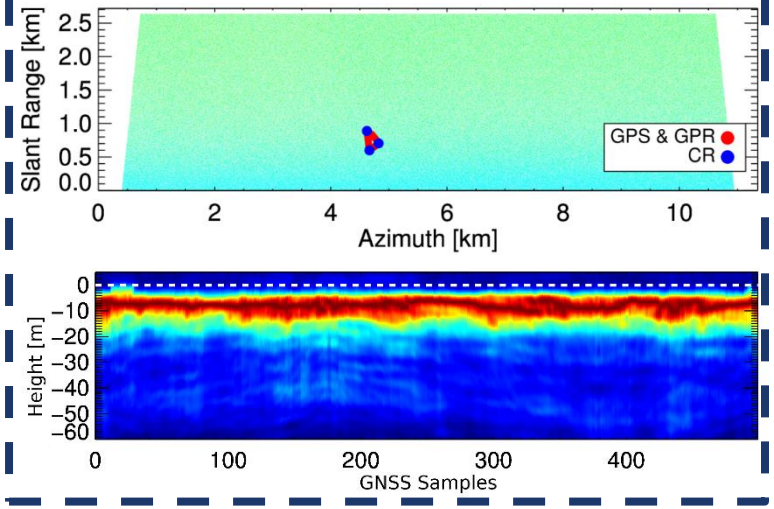
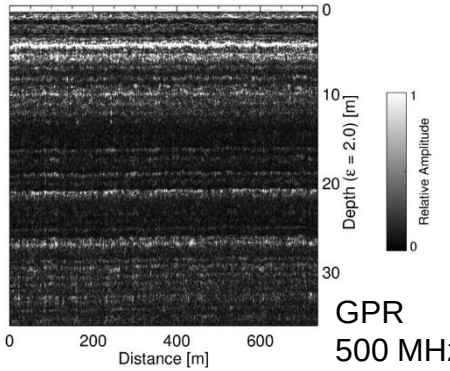


South Dome

Less melting
Distinct layers
Lower density



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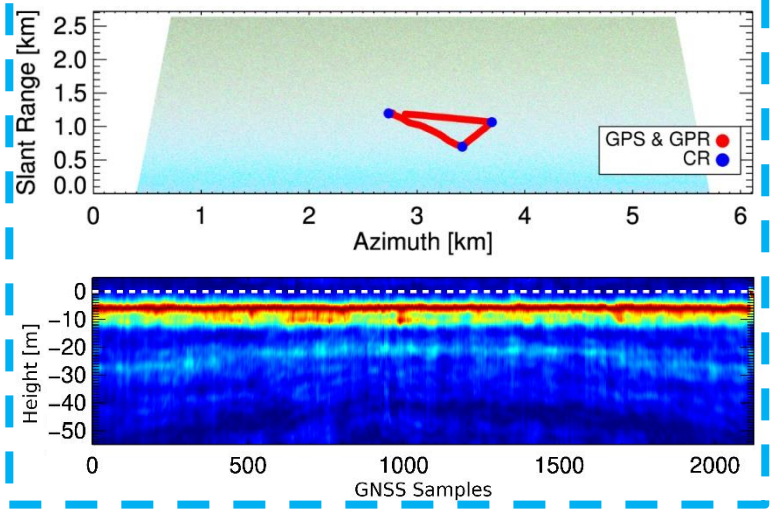
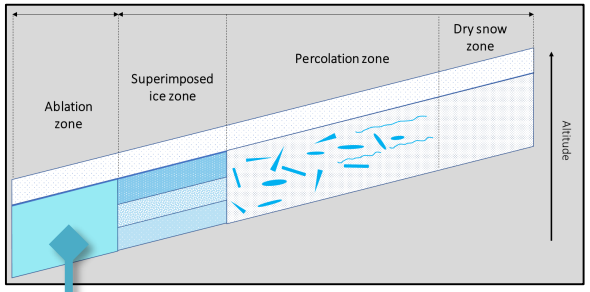
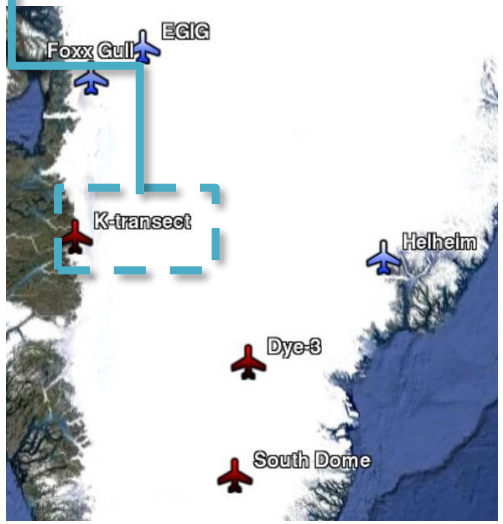
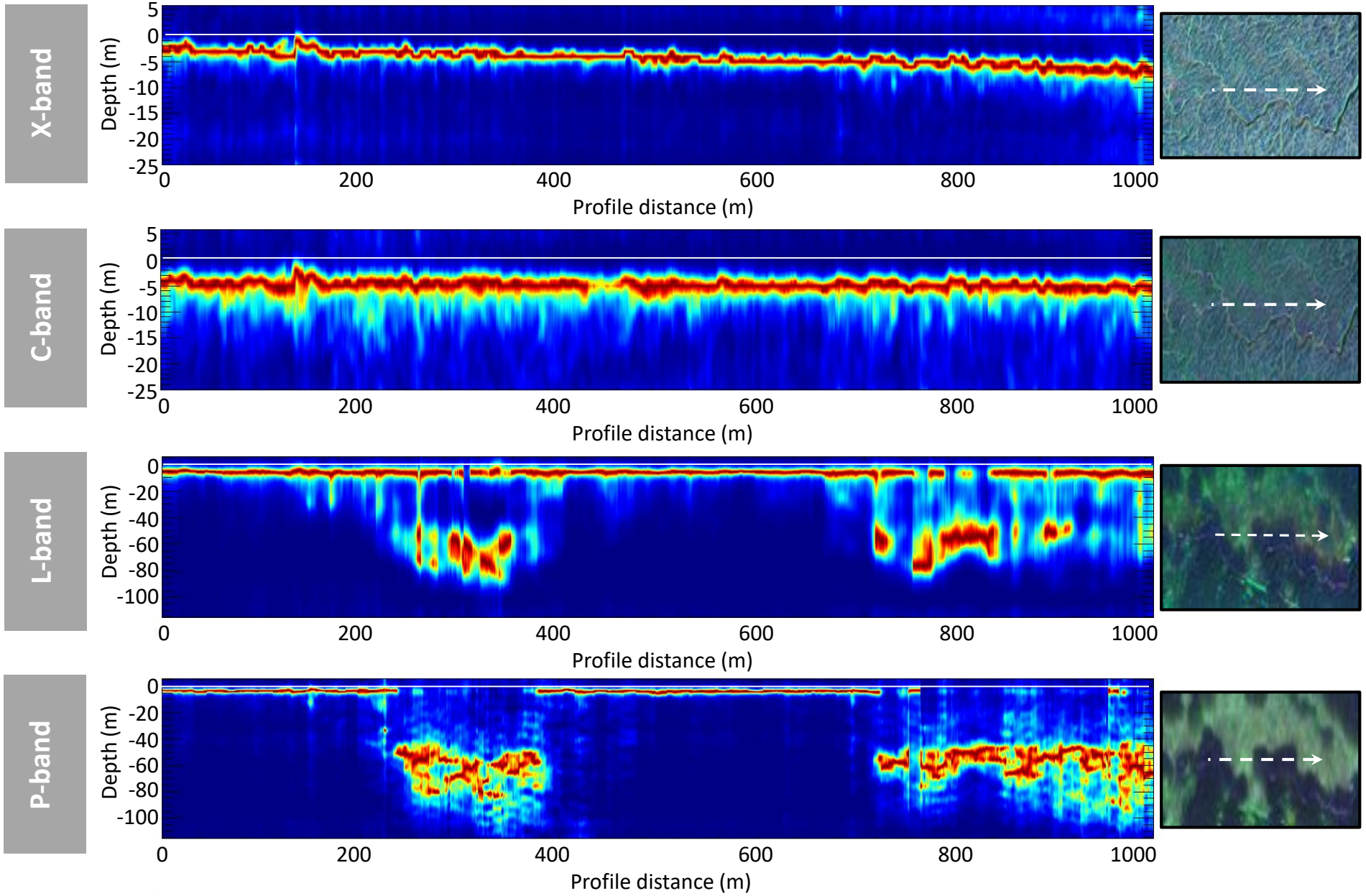


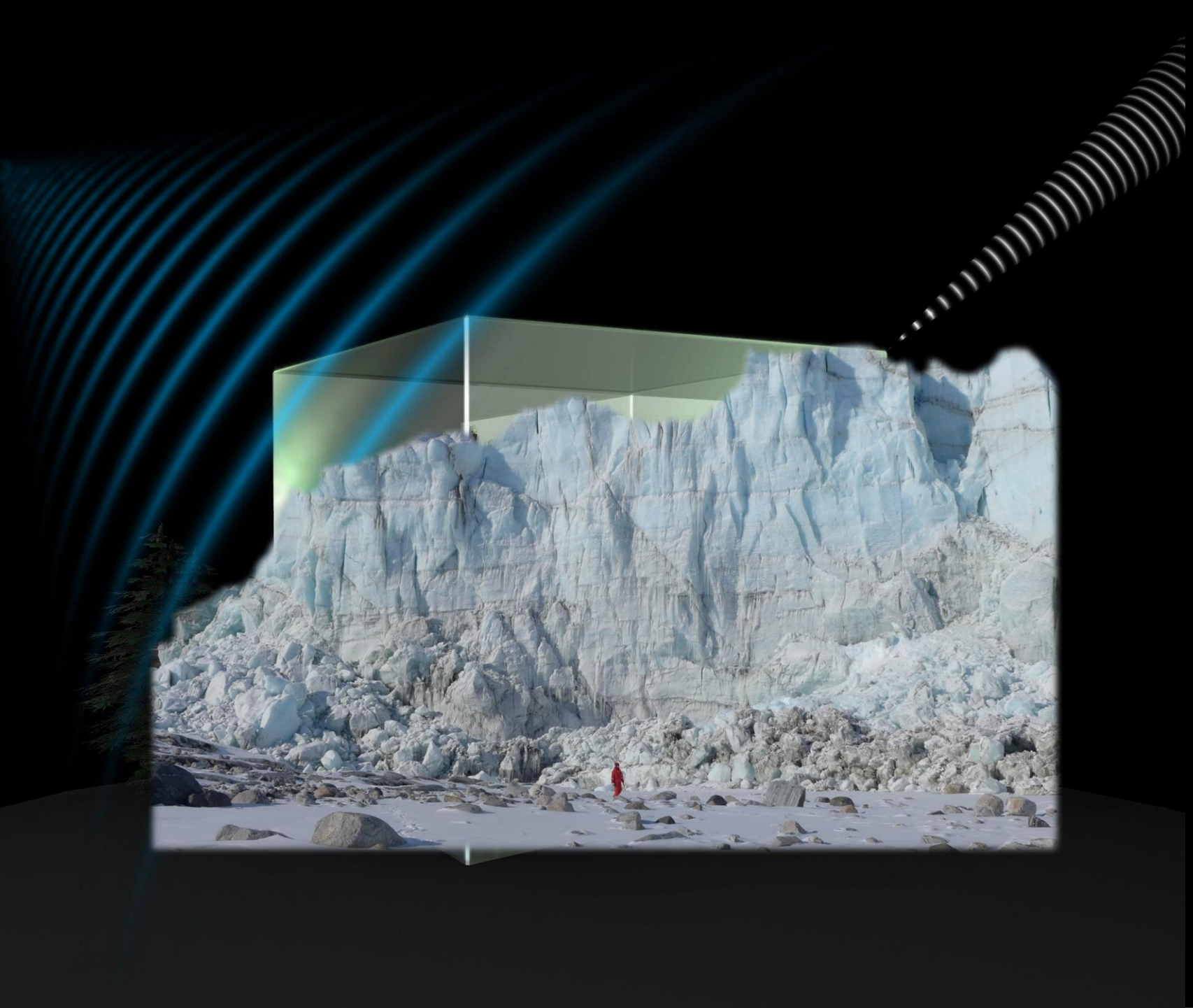
Why P-band TomoSAR?







DLR



Why Land Ice Structure?

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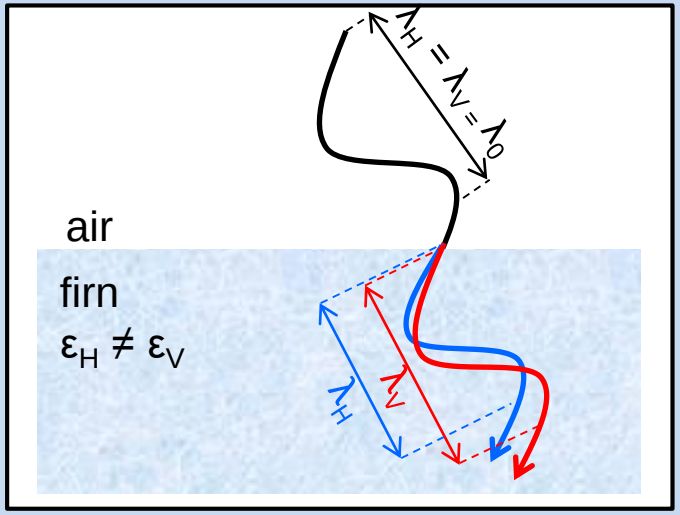
Experimental Outlook: Firn Thickness with PolSAR Modelling

- Firn → temperature gradient **metamorphism**
  → vertical elongation of grains
- $\varepsilon_H \neq \varepsilon_V$ → Different propagation for *H* and *V* waves

Phase difference between HH and VV channels (CPD)

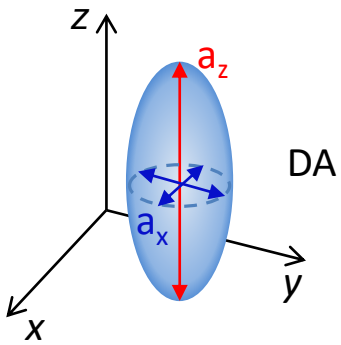


PolSAR Model:

$$CPD = \phi_{HH} - \phi_{VV} = f(\boldsymbol{\sigma}(\boldsymbol{z}), \boldsymbol{\rho}, \boldsymbol{DA})$$

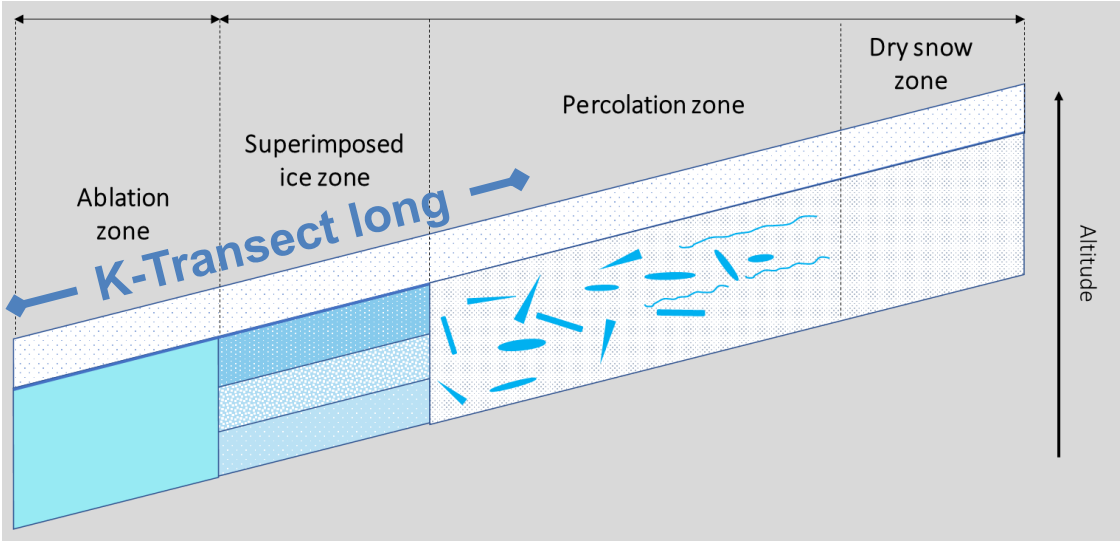
Parameters:

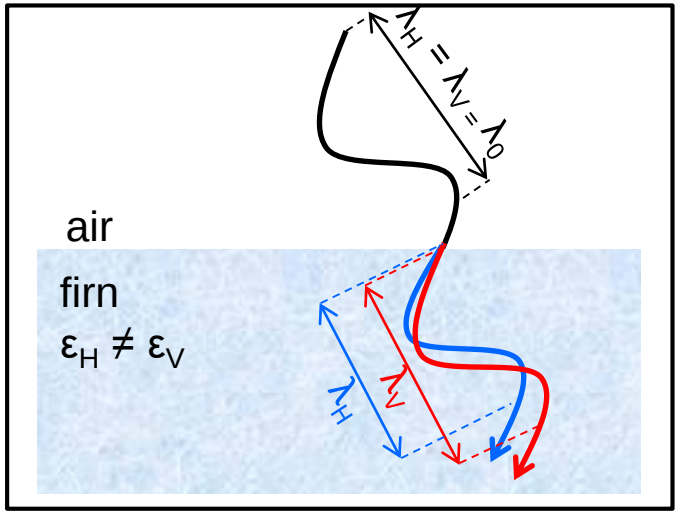
- $\boldsymbol{\sigma}(\boldsymbol{z})$ scattering structure
- **ρ** Density
- **DA** Degree of Anisotropy



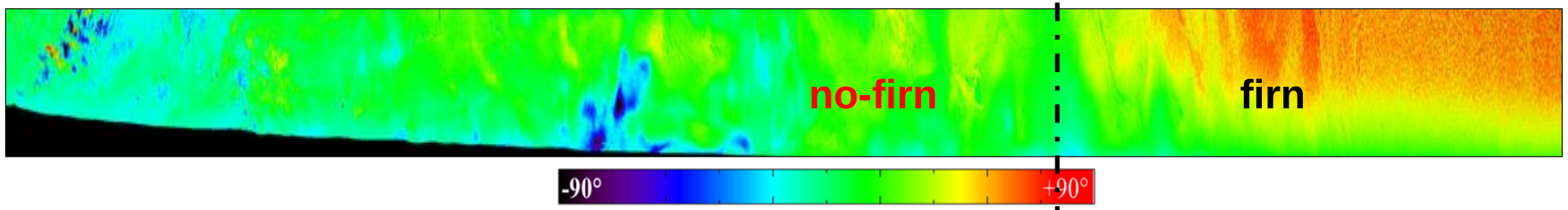
DA = a_z/a_x Degree of Anisotropy

Experimental Outlook: Firn Thickness with PolSAR Modelling

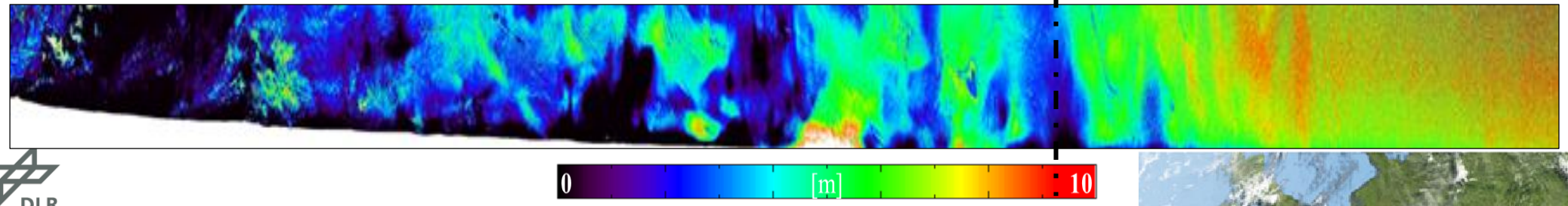




L-band CPD $(\phi_{HH} - \phi_{VV})$

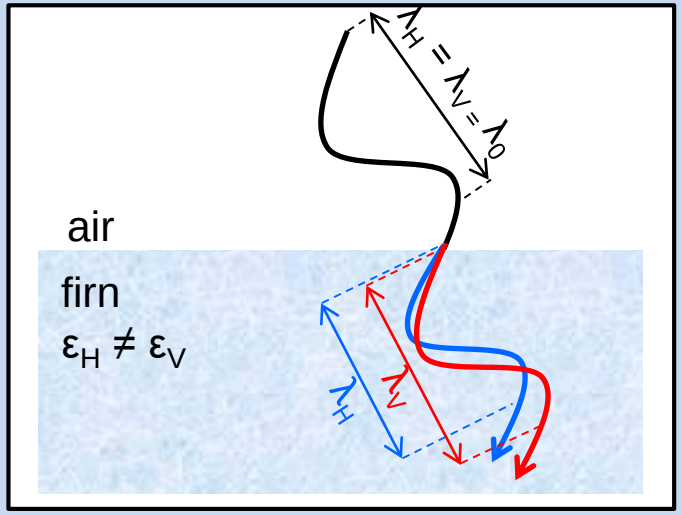


PolSAR Model: Firn Thickness Estimation



Experimental Outlook: Firn Density from PolSAR and TomoSAR

PolSAR CPD Model

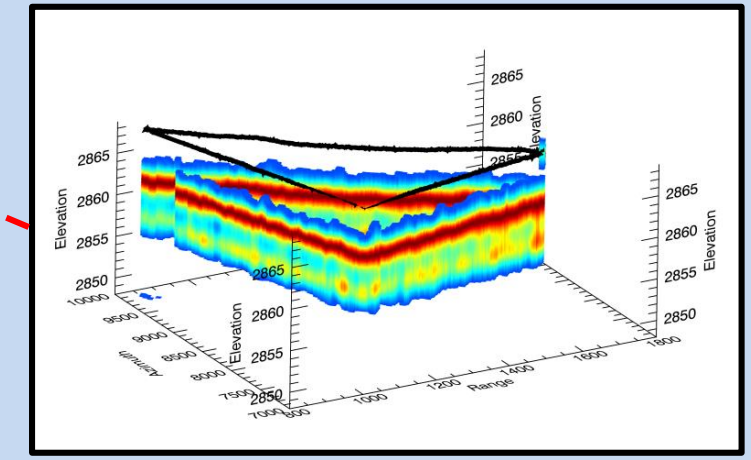


$$CPD = \phi_{HH} - \phi_{VV}$$
$$= f\,(\boldsymbol{\sigma(z)}, \boldsymbol{DA}, \boldsymbol{\rho})$$

Multiple incidence angles

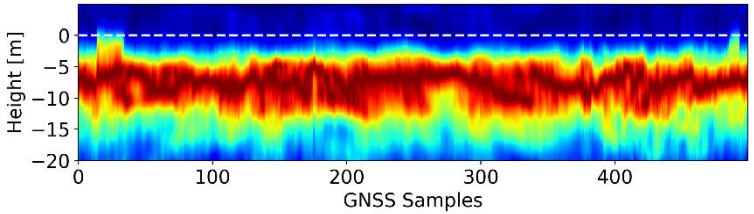
Solving for a bulk density $\boldsymbol{\rho}$

TomoSAR Scattering Structure

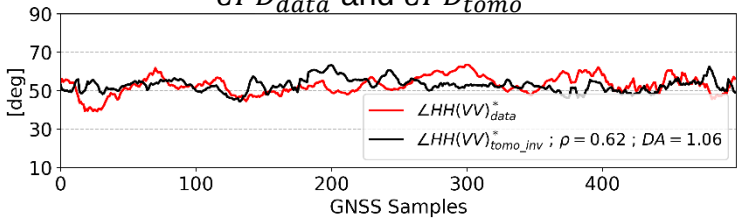


Experimental Outlook: Firn Density from PolSAR and TomoSAR

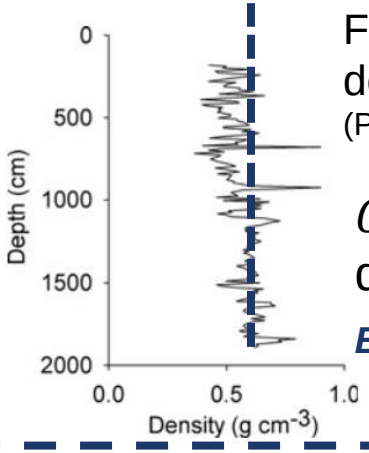
EGIG T05



CPD_{data} and CPD_{tomo}



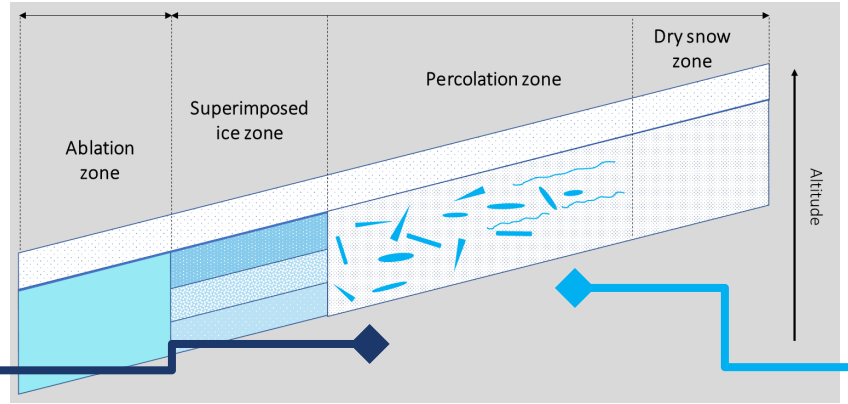
$\rho = 0.62\ g/cm^3$
$DA = 1.06$



Firn core density
(Parry et al., 2007)

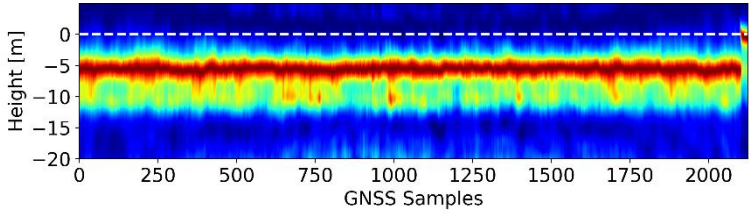
CPD_{tomo} density

EGIG T05



Firn Density from Space?

South Dome



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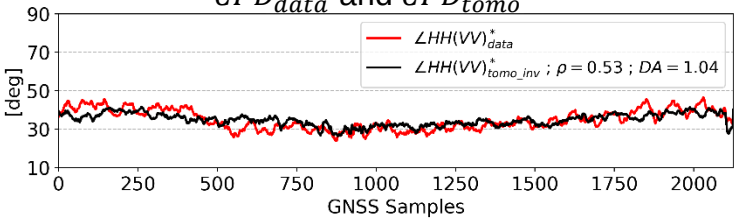

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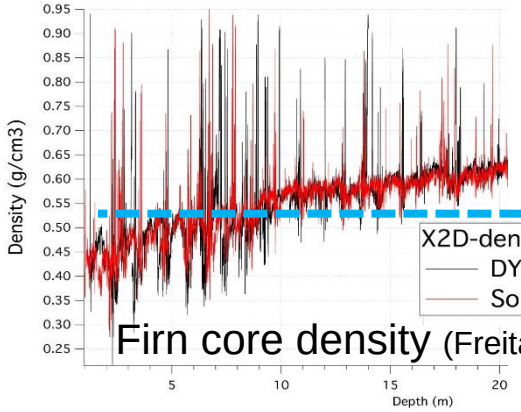
$DA = 1.04$



Firn core density (Freitag et al., 2014)

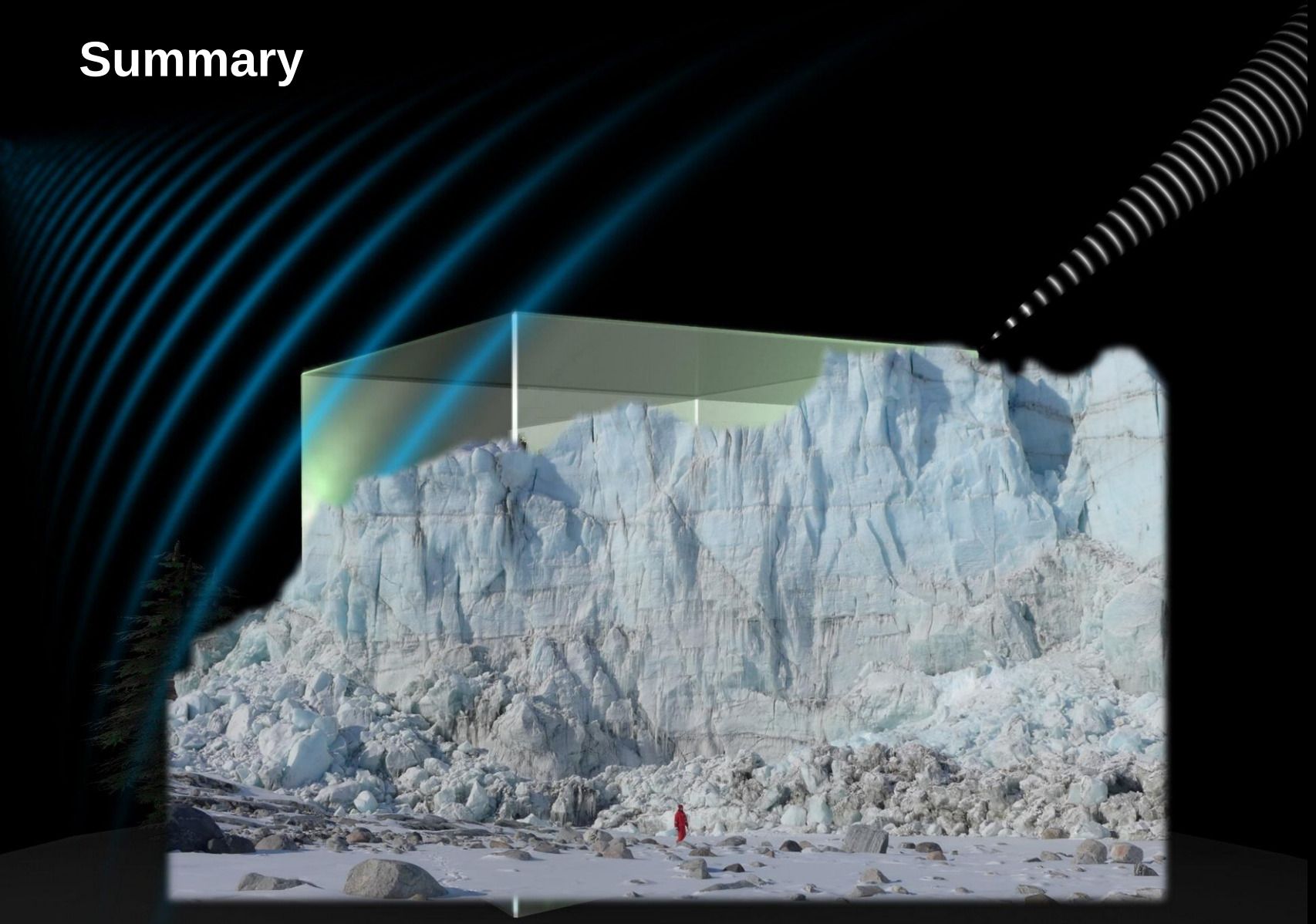
CPD_{tomo} density

South Dome

Parry *et al.*, "Investigations of meltwater refreezing and density variations in the snowpack and firn thin the percolation zone of the Greenland ice sheet," *Annals of Glaciology*, vol. 46, pp. 61–68, 2007.

J. Freitag, S. Kipfstuhl, S. Hoerz, L. Eling, B. Vinther, and T. Popp, "Melt layer statistic of two firn cores recently drilled at Dye3 and South Dome in the dry snow zone of Southern Greenland," *presented at the EGU General Assembly*, Vienna, Austria, Apr./May 2014.

Summary



P-band Penetration

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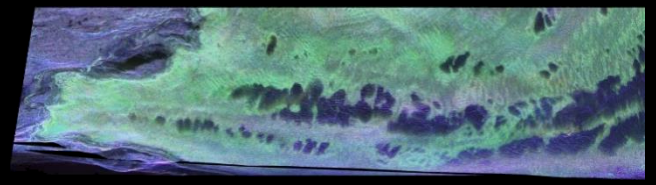
Polarimetry



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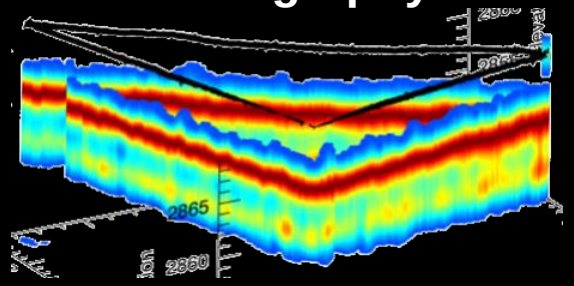
Tomography



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Land Ice Structure Information with BIOMASS